The Error Theorem

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Abstract

This text is a small experiment between literature and mathematics. It begins with an absurd scene, moves through a multiple-choice problem, and ends with a reflection on probability, symmetry, and the difference between real information and imaginary hope. The result is a playful theorem about error: how it deceives us, how it seduces us, and why it cannot be treated as a source of certainty.

1 Introduction

Imagine.

One morning, after waking from troubled dreams and still trapped in bed, Sregor Gamsa discovered that he had been transformed into a monstrous insect. Lying on his back, he looked at the clock ticking on the dresser and felt, with the clarity of a mathematical punishment, that he was already late.

But that was not the worst of his problems.

His real problem was different: that morning there would be a multiple-choice exam in Statistics VI.

He tried to move. He could not. He tried to convince himself that the world might show some compassion toward the metamorphosed. That also failed.

So, like any man who realizes that time and logic are against him, Sregor did what remained to a being cornered between absurdity and urgency: he thought.

"Maybe I can still make it there with ten minutes left."

The idea was ridiculous, but no more ridiculous than the body he now carried. With ten minutes left, he could guess all the questions. And then came the truly dangerous part: not despair, but hope.

"If I am going to get almost everything wrong by chance, maybe I can find a way to be wrong more intelligently."

He stared at the ceiling as if staring at a cursed system of equations. The first intuition was simple: in a question with four alternatives, if I know nothing, my chance of success is only $\frac{1}{4}$. Therefore, if I can eliminate one option, my chance rises to $\frac{1}{3}$.

But the human mind is a strange machine. It does not content itself with noticing that chance improves when there is information.

"What if I could use error as a stepping stone?" thought Sregor's mind.

Thus was born, or perhaps merely revealed, what we call here the *Error Theorem*.

2 The Problem

Consider a multiple-choice question with n alternatives, exactly one of which is correct.

If no additional information is given, and if we assume symmetry among the alternatives, then the chance of answering correctly on a single guess is

$$P(\text{correct}) = \frac{1}{n}.$$

In the particular case $n = 4$, we obtain

$$P(\text{correct}) = \frac{1}{4}.$$

So far, nothing strange.

The problem arises when Sregor imagines the following procedure:

1. choose one alternative at random;
2. eliminate that alternative;
3. choose again, at random, one alternative among the remaining ones.

At first glance, this seems advantageous. After all, if one alternative has been eliminated, the new problem now has only $n-1$ options, and so the probability of success on the second guess would seem to be

$$P(\text{correct}) = \frac{1}{n-1}.$$

For $n = 4$, this gives the impression of an improvement from $\frac{1}{4}$ to $\frac{1}{3}$.

But this is exactly where intuition betrays us.

The essential point is that the reduction from n to $n-1$ alternatives did not come from real information about the correct answer. It came from a previous random choice.

In other words, the problem was not transformed into two independent problems; it remains a single probabilistic experiment with two stages.

To make this explicit, define the events:

- E_1: the alternative chosen on the first guess is wrong;
- C_1: the alternative chosen on the first guess is correct;
- C_2: the alternative chosen on the second guess is correct.

The central point is the following: the second guess can only be analyzed conditional on the result of the first.

If the first guess is wrong, then $n-1$ alternatives remain, among which exactly one is correct. Therefore,

$$P(C_2 \mid E_1) = \frac{1}{n-1}.$$

On the other hand, if the first guess is correct, then the correct alternative has been eliminated. In that case, the second guess can no longer recover the success, and therefore

$$P(C_2 \mid C_1) = 0.$$

Now apply the law of total probability:

$$P(C_2) = P(E_1)\, P(C_2 \mid E_1) + P(C_1)\, P(C_2 \mid C_1).$$

Since

$$P(E_1) = \frac{n-1}{n} \quad \text{and} \quad P(C_1) = \frac{1}{n},$$

we have

$$P(C_2) = \frac{n-1}{n} \cdot \frac{1}{n-1} + \frac{1}{n} \cdot 0.$$

Therefore,

$$P(C_2) = \frac{1}{n}.$$

In the case $n = 4$,

$$P(C_2) = \frac{3}{4} \cdot \frac{1}{3} + \frac{1}{4} \cdot 0 = \frac{1}{4}.$$

Thus, the probability of answering correctly on the second guess is exactly the same as if we had stopped at the initial guess.

The apparent advantage of moving from $\frac{1}{4}$ to $\frac{1}{3}$ exists only within the subset of cases in which the first guess was wrong. But that subset does not cover all possible outcomes.

In fact, there is a probability of $\frac{1}{4}$ that the first guess eliminates precisely the correct alternative, making success impossible. This case cancels the apparent gain in the other cases.

We can see this by listing the scenarios explicitly, for $n = 4$.

Suppose the alternatives are $\{A, B, C, D\}$ and that only A is correct.

- If the first guess chooses A, with probability $\frac{1}{4}$, then the correct answer has been eliminated.
- If the first guess chooses B, with probability $\frac{1}{4}$, then $\{A, C, D\}$ remain, and the chance of being correct on the second guess is $\frac{1}{3}$.
- If the first guess chooses C, with probability $\frac{1}{4}$, then $\{A, B, D\}$ remain, and the chance of being correct on the second guess is $\frac{1}{3}$.

- If the first guess chooses D, with probability $\frac{1}{4}$, then $\{A, B, C\}$ remain, and the chance of being correct on the second guess is $\frac{1}{3}$.

Summing the favorable cases for the final success, we obtain

$$P(C_2) = 3 \cdot \left(\frac{1}{3} \cdot \frac{1}{4}\right) = \frac{1}{4}.$$

What seemed like a trick for increasing the chance of success was, in fact, only a rewriting of the same random experiment.

The second guess seems more promising only because it is viewed under the condition that the first one has already failed. But this condition does not arise from extra information about the problem; it arises from chance itself.

In other words, the system did not become smaller because it knew more.

It became smaller because a random choice had already been made.

And that is exactly where intuition fails:

Seeming to reduce the problem is not the same as reducing uncertainty.

3 The Error Theorem

Theorem 1 (Error Theorem)**.** *In a multiple-choice problem with n equiprobable alternatives, eliminating one option changes the probability of success if and only if that elimination is informative, that is, if it carries some correlation with the correct answer.*

If the elimination occurs at random, or without any knowledge of which alternative is true, then there is no real probabilistic gain: the total probability of success remains

$$P = \frac{1}{n}.$$

In particular, the apparent transition from n to $n-1$ alternatives does not, by itself, constitute a reduction of uncertainty; it does so only when it introduces new information about the problem.

This theorem is simple in form, but subtle in content.

The mind tends to confuse two distinct situations. In the first, an alternative is eliminated because something has been learned about it; in that case, the space of possibilities really is reduced. In the second, the elimination occurs only as a consequence of a previous guess; in that case, the problem was not truly simplified, but merely reorganized into conditioned stages.

The difference between these two situations is fundamental.

In the first case, there is information. In the second, there is only chance rewritten.

For this reason, the real point of the Error Theorem is not that "making mistakes helps," but rather that the appearance of improvement may arise when we look only at part of the experiment and forget the complete process.

The first operation can change the mathematics. The second changes only the narrative.

4 A Formal View

Let $\Omega = \{1, 2, \ldots, n\}$ be the set of possible alternatives, and let I be the random variable that indicates the correct alternative. We assume a uniform prior distribution:

$$P(I = i) = \frac{1}{n}, \qquad i \in \Omega.$$

Now consider an event E associated with an elimination observed in the choice process. The central question is whether this event carries information about I.

If E is *informative*, that is, if it implies that a given alternative A_k is false, then the set of possibilities becomes $\Omega \setminus \{k\}$. By symmetry among the remaining alternatives, we obtain

$$P(I = i \mid E) = \frac{1}{n-1}, \qquad i \neq k.$$

On the other hand, if E is statistically independent of I, then the elimination event does not alter the prior distribution:

$$P(I = i \mid E) = P(I = i) = \frac{1}{n}.$$

Proposition 1. *Eliminating an alternative changes the probability of success if and only if it introduces information about the correct alternative.*

Proof. First, suppose that E is the event of certainly eliminating alternative A_k. Then

$$P(I = k \mid E) = 0.$$

Since the remaining alternatives are symmetric, we have

$$P(I = i \mid E) = P(I = j \mid E), \qquad i, j \neq k.$$

Denoting this common value by q, it follows that

$$\sum_{i \neq k} P(I = i \mid E) = 1 \quad \Longrightarrow \quad (n-1)q = 1.$$

Therefore,

$$P(I = i \mid E) = \frac{1}{n-1}, \qquad i \neq k.$$

Now suppose that E is independent of I. By Bayes' formula,

$$P(I = i \mid E) = \frac{P(E \mid I = i)\, P(I = i)}{P(E)}.$$

Since $E \perp I$, we have

$$P(E \mid I = i) = P(E), \qquad \forall i \in \Omega.$$

Substituting,

$$P(I = i \mid E) = \frac{P(E) \cdot \frac{1}{n}}{P(E)} = \frac{1}{n}.$$

Therefore, when E is informative, the posterior distribution changes to $\frac{1}{n-1}$ among the remaining alternatives; when E is independent, the distribution remains uniform at $\frac{1}{n}$. $\square$

5 A Quantum Information Perspective

The conceptual structure of the Error Theorem finds a natural parallel in Quantum Information Theory.

In quantum mechanics, the state of a system is described by a vector $|\psi\rangle$ in a Hilbert space. The uncertainty associated with this state is not merely a practical limitation, but a fundamental property of the system.

Consider a quantum system prepared in a uniform superposition of n orthonormal states:

$$|\psi\rangle = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} |i\rangle \,.$$

The probability of obtaining outcome i when measuring in the basis $\{|i\rangle\}$ is

$$P(i) = \frac{1}{n},$$

in perfect analogy with the classical multiple-choice problem under symmetry.

Now suppose we perform an operation on the system that eliminates one of the possible states.

The fundamental question is: *does this elimination correspond to an informative measurement?*

If the operation is a projective measurement that certainly discards state $|k\rangle$, then the system collapses to the subspace generated by the remaining states. The new state is given by

$$|\psi'\rangle = \frac{1}{\sqrt{n-1}} \sum_{i \neq k} |i\rangle \,,$$

and the probability distribution becomes

$$P(i \mid \text{elimination of } k) = \frac{1}{n-1}, \qquad i \neq k.$$

In this case, there is a real reduction in uncertainty: the measurement extracted information about the system.

On the other hand, consider an operation that removes a state independently of the content of the system — for example, a random loss or a filtering process that does not depend on the correct basis.

In that case, the operation is described by a quantum channel that is not correlated with the observable of interest. The effective density matrix of the system, after conditioning on the success of the operation, contains no additional information about which state was correct.

Analogously to the classical case, the relevant probability distribution remains unchanged:

$$P(i \mid E) = \frac{1}{n}.$$

The central distinction can be expressed in informational terms:

- If the operation has nonzero **mutual information** with the measured observable, then it reduces the entropy of the system.
- If the operation is **independent**, then it does not alter the relevant entropy, even if the apparent state space becomes smaller.

This point is deeply nonclassical.

In quantum mechanics, not every interaction produces knowledge. A system may be disturbed, filtered, or even partially destroyed without any useful information being extracted about the variable of interest.

Thus, the quantum analogue of the Error Theorem may be formulated as:

> Not every reduction of the state space reduces uncertainty about an observable.

Or, in more direct terms:

> It is not the transformation of the system that generates information, but the correlation between that transformation and what one wishes to know.

This is exactly the same logical structure observed earlier.

In Sregor Gamsa's problem, one alternative is eliminated without any knowledge of the correct answer. The system seems smaller, but the uncertainty remains.

In the quantum setting, a state may be removed or a subspace discarded without implying any gain of information about the relevant observable.

In both cases, the error lies in confusing:

- **structural reduction of the space**, with
- **informational reduction of uncertainty**.

Quantum mechanics merely makes this distinction unavoidable.

It forces us to recognize that information is not a property of what we do to the system, but of what we are able to infer from it.